

Step 4 — Calibration and Cointegration

Step 4 — Calibrate the basket to a probability, then test for cointegration

This report consolidates the four Step 4 text outputs (two-point calibration and the cointegration gate, at both the high-frequency and daily grains) into one document, and embeds the two paired figures.

Step 3 produced B_t , the basket’s buy-and-hold level in return units. Step 4 maps B_t onto an implied probability using two known points, then tests whether that implied probability and the Kalshi PRES–2024–DJT price are cointegrated — the precondition for any VECM / Hasbrouck price-discovery analysis.

- **high_frequency** — 1-minute bars, overnight gaps included (9,325 bars, 24 days).
- **daily** — one close-to-close observation per day, which naturally carries the overnight move (24 days).

Part 1 — Two-point calibration

Anchors (same at both grains):

- **p0 (sample start):** $P(\text{Trump}) = 0.447$ — Silver Bulletin model update, 2024-09-30/10-01 (nearest update to the 2024-10-04 start).
- **p1 (election night):** $P(\text{Trump}) = 1.0$ — Trump won.

$$\text{implied_p_t} = q_0 + (B_t - B_{@p0}) * (q_1 - q_0) / (B_{@p1} - B_{@p0})$$

$B_{@p0}$ is the first-bar level (= 0 by construction); $B_{@p1}$ is the election-night level. The **slope** is probability points per unit basket return.

high_frequency

plan	B_terminal	slope	impl_p min	impl_p max
A_sign_equal	+0.0917	+6.03	−0.742	+1.111
B_beta_weighted	+0.1546	+3.58	+0.115	+1.028
C_min_variance	−0.0276	−20.04	−0.988	+4.731

daily

plan	B_terminal	slope	impl_p min	impl_p max
A_sign_equal	+0.0870	+6.36	−0.605	+1.000
B_beta_weighted	+0.1568	+3.53	+0.208	+1.000
C_min_variance	+0.0035	+156.79	−26.002	+15.290

Reading the three plans

- **Plan B (beta-weighted) is the best-behaved at both grains.** Its calibrated probability stays essentially inside $[0,1]$ — $[+0.115, +1.028]$ high-frequency and a clean $[+0.208, +1.000]$ daily. This is the plan to carry downstream.
- **Plan A (sign-equal)** calibrates with a positive slope but is too noisy: it overshoots to -0.74 (high-frequency) / -0.61 (daily), so it cannot be read as a clean probability.
- **Plan C (min-variance) is unusable at both grains, but for different reasons.**
 - *high_frequency*: the basket level actually **fell** over the window ($B_terminal = -0.028$) while $P(\text{Trump})$ rose, so the slope is **negative** (-20.0) — an inverted signal that goes down when Trump's odds go up.
 - *daily*: the terminal level is **almost exactly zero** ($+0.0035$), so the slope **explodes** ($+156.8$) and the calibrated series blows up to ± 26 . Either way plan C is not a probability. It matches its Step-2 out-of-sample weakness: the min-variance weights are best in-sample but do not hold up out-of-sample. Kept in the CSVs for transparency only.

The reference is Silver's ~Oct 1 update while the basket starts Oct 4; the ~3-day offset is the nearest model update to the sample start.

Part 2b — Cointegration gate

Before any VECM / Hasbrouck, test whether each calibrated basket and Kalshi share one efficient price. If they do, the spread is stationary (mean-reverting). If the spread wanders (a unit root), the pair is **not** cointegrated and the Hasbrouck information-share machinery is not legitimate for it.

Three readings per plan:

1. **spread(1,−1) ADF** — ADF on $\text{implied_p} - \text{kalshi}$; this is what the two-point calibration targets (both are meant to be $P(\text{Trump})$).
2. **Engle-Granger** — regress implied_p on kalshi , ADF on the residual (lets the data pick the slope instead of imposing 1).
3. **Johansen trace** — $r = 0$ vs $r \geq 1$; trace stat vs its 95% critical value.

Cointegrated requires ADF/EG $p < 0.05$, or Johansen trace above its 95% value.

high_frequency

Full window 2024-10-04..11-06 — n = 9,325, 24 days, Kalshi fresh 75%

plan	spread(1,-1) ADF p	EG b	EG ADF p	Johansen trace (95%=15.49)	verdict
A_sign_equal	0.461	-0.01	0.534	3.31	no
B_beta_weighted	0.499	+0.54	0.653	2.98	no
C_min_variance	0.503	+4.49	0.665	4.42	no

High-liquidity sub-window 2024-10-21..11-06 — n = 5,051, 13 days, Kalshi fresh 95%

plan	spread(1,-1) ADF p	EG b	EG ADF p	Johansen trace (95%=15.49)	verdict
A_sign_equal	0.797	+2.56	0.383	4.67	no
B_beta_weighted	0.794	+1.44	0.587	3.21	no
C_min_variance	0.681	-4.15	0.377	7.32	no

daily

Full window 2024-10-04..11-06 — n = 24, 24 days, Kalshi fresh 100%

plan	spread(1,-1) ADF p	EG b	EG ADF p	Johansen trace (95%=15.49)	verdict
A_sign_equal	0.420	+0.17	0.259	13.80	no (near)
B_beta_weighted	0.095	+0.61	0.788	7.46	no (ADF marginal)
C_min_variance	0.306	-30.80	0.061	5.34	no

High-liquidity sub-window 2024-10-21..11-06 — n = 13, 13 days, Kalshi fresh 100%

plan	spread(1,-1) ADF p	EG b	EG ADF p	Johansen trace (95%=15.49)	verdict
A_sign_equal	0.128	+2.72	0.603	13.29	no
B_beta_weighted	0.587	+1.46	0.664	22.82	COINTEGRATED*
C_min_variance	0.989	+36.29	0.973	16.51	COINTEGRATED*

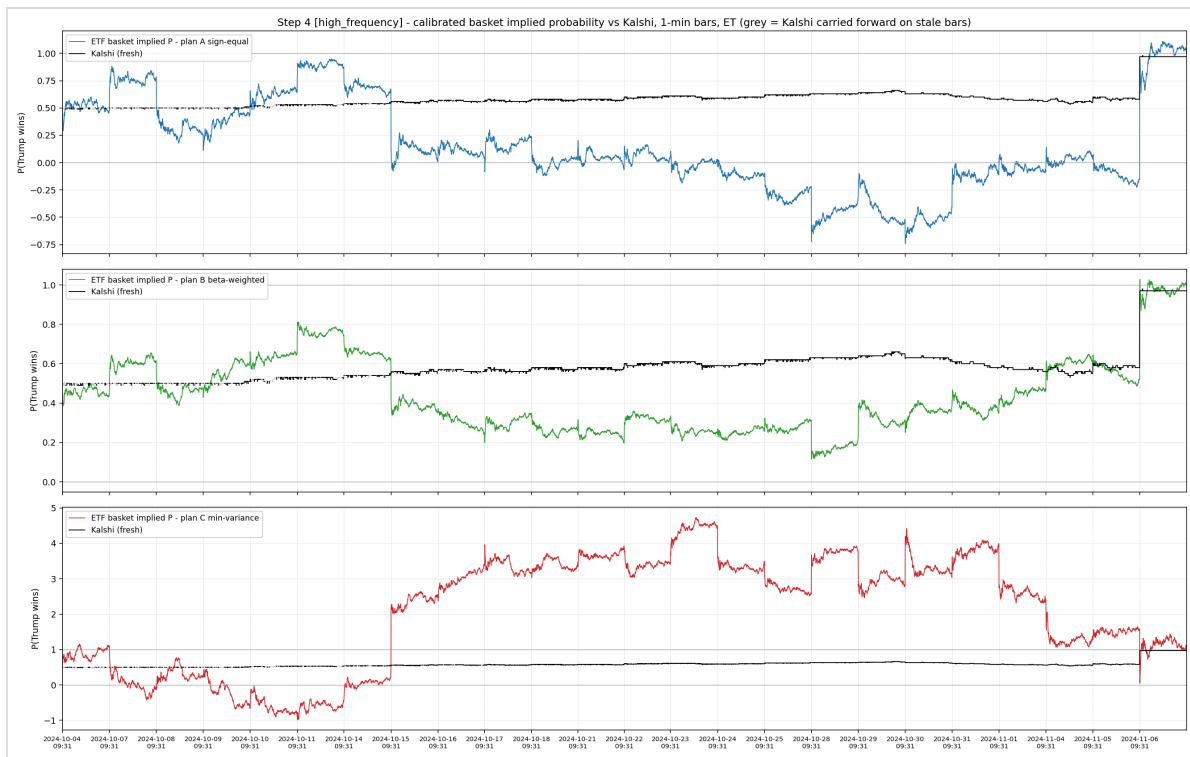
* **These two flags are not trustworthy.** At n = 13 the Johansen trace test uses asymptotic critical values that over-reject in small samples, so trace > 15.49 is not credible here; for the same plan B the

ADF (0.587) and Engle–Granger (0.664) both say *no*, and the plots show the basket and Kalshi diverging between the anchors. Plan C is degenerate anyway. Treat both as small-sample artifacts.

Figures

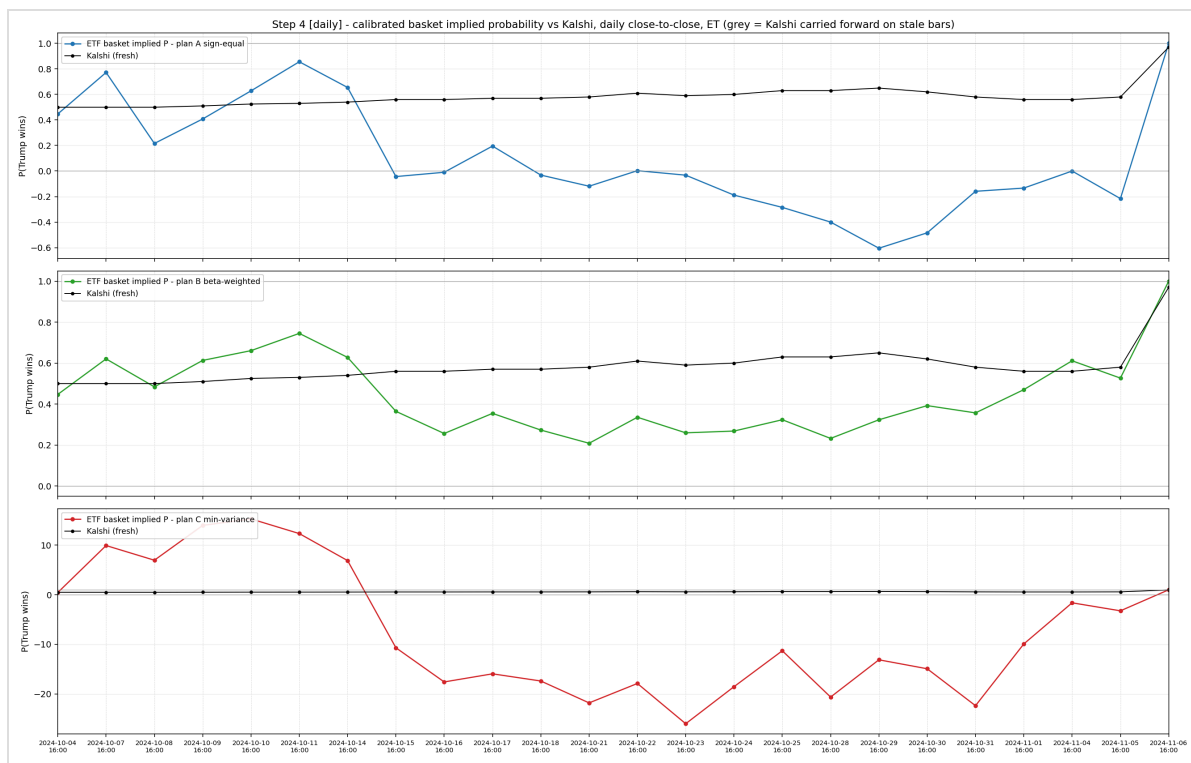
Calibrated basket implied probability vs Kalshi, one weight plan per panel. Each panel has its own y-axis (plan A and especially plan C stray outside [0,1]). Grey = Kalshi carried forward on stale bars; black = fresh Kalshi trade.

high_frequency (1-minute bars)



Step 4 pair — high frequency

daily (close-to-close)



Step 4 pair — daily

Bottom line

The two-point calibration forces implied_p and Kalshi to agree **only** at the two anchors (sample start and election night). Between them the basket is driven by moves the market does not share, so the spread wanders even though the endpoints match.

- **No robust cointegration at either grain.** Every ADF and Engle–Granger test says *no*; the only two “COINTEGRATED” flags (daily plan B/C sub-window Johansen) are small-sample artifacts contradicted by the other tests and the figures.
- **Daily is better-behaved but does not rescue the relationship.** Plan B stays in a valid probability range and its full-window spread ADF is marginal ($p = 0.095$), but the basket still diverges from Kalshi between anchors.
- **This is not a weight-tuning problem.** Plan B is already the beta-weighted election-mimicking basket; any linear combination of these 11 broad sector ETFs carries too much non-election variance, which accumulates in the level and breaks cointegration. Contemporaneous return co-movement exists (mostly overnight) but does not amount to a shared efficient price.

Because cointegration fails, a VECM / Hasbrouck information-share analysis on this pair is not legitimate. The honest conclusion is that the sector-ETF basket and the Kalshi election price do **not** share one common price to attribute discovery to.